

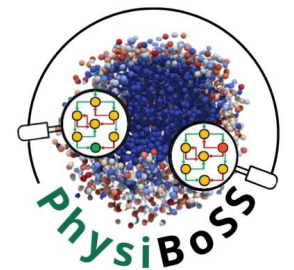
Computational Biology in the LLM era:

coding, tools, workflows, and agents.

Marco Ruscone, PhD

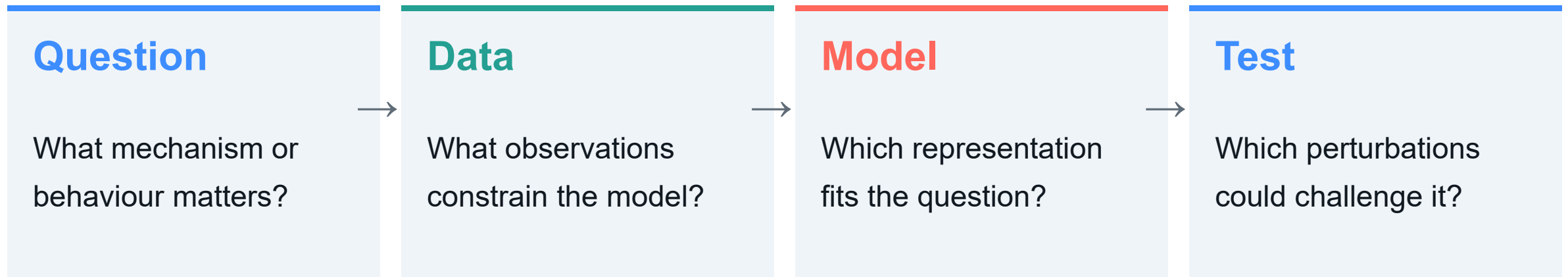
Elliott Jacopin, PhD

02/10/2026



Biological questions become computational workflows

From cellular mechanisms to tissue dynamics, research connects questions, evidence, models and tests.



Inspect results → refine the question, data or model

Models connected to biological data can contribute to digital twins.

Hybrid modelling connects different representations

Topology, intracellular dynamics and cell populations require different tools—and explicit interfaces.



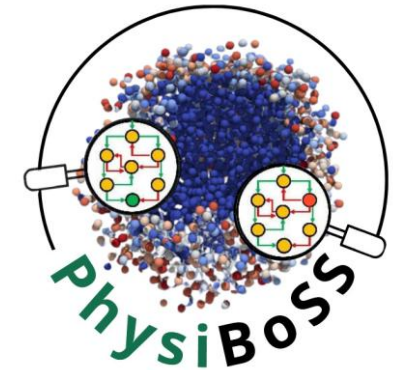
NeKo

Prior knowledge → topology



MaBoSS

Boolean rules → dynamics



PhysiBoSS

Intracellular states → cells

One scientific question · coordinated specialists · a reproducible record

Before: the researcher carried every handoff

Pipelines automate execution and data handling; the researcher still connects tools, assumptions and results.

01 NeKo

Construct topology
Inspect paths and signs

02 Literature

Check references
Resolve context and direction

03 Boolean rules

Export BNET
Inspect and repair logic

04 MaBoSS

Run WT and mutants
Diagnose failed behaviour

05 PhysiBoSS

Map states to phenotypes
Configure and run cellular
simulations

06 Reproducibility

Save versions and settings
Write the reports, thesis etc..

Every revision meant revisiting code, papers, assumptions and files.

From writing scripts to describing an objective

Coding agents can translate natural-language requests into scripts, tool operations and visualizations.

Describe

State the question, data and expected outputs.

Implement

The agent execute code and operates tools.

Inspect

Review the result and request a bounded change.

Natural language can become the interface to an executable research workflow.

Skills explain the procedure; MCP exposes the tools

Skills package task procedures. The Model Context Protocol (MCP) connects agents to tools and data.

Skills

Procedures, domain context and checks

Example: inspect the approved model;
run matched WT and perturbation analyses.

MCP servers

Callable software and data access

Example: query a network, run a simulation,
retrieve literature or export artifacts.

Article | [Open access](#) | Published: 15 June 2026

**Intelligent tool orchestration for rapid mechanistic model
prototyping: MCP servers as AI-biology interfaces**

[Marco Ruscone](#) , [Miguel Vazquez](#) & [Alfonso Valencia](#)

[npj Systems Biology and Applications](#) **12**, Article number: 145 (2026) | [Cite this article](#)

The agent combines the instructions with available tools, then reports the outcome.

Agent definitions guide; software controls and checks

Roles and instructions influence behaviour. Execution boundaries and validation need explicit mechanisms.

Definitions

Role, scope and responsibilities
Skills and expected outputs

Execution

Tool access and permissions
Bounded tasks and handoff
checks

Supervision

Evaluate scientific assumptions
Approve changes and
conclusions

A well-written agent definition is guidance, not a guarantee of correct behaviour.

Now: prompt, inspect, decide and keep the record

The researcher states the objective; the system coordinates the work and surfaces scientific choices.

A modelling objective

**Build a toy ER+ cell-cycle model.
Test cyclin order and arrest after
ESR1 loss or CDK4/6 inhibition.**

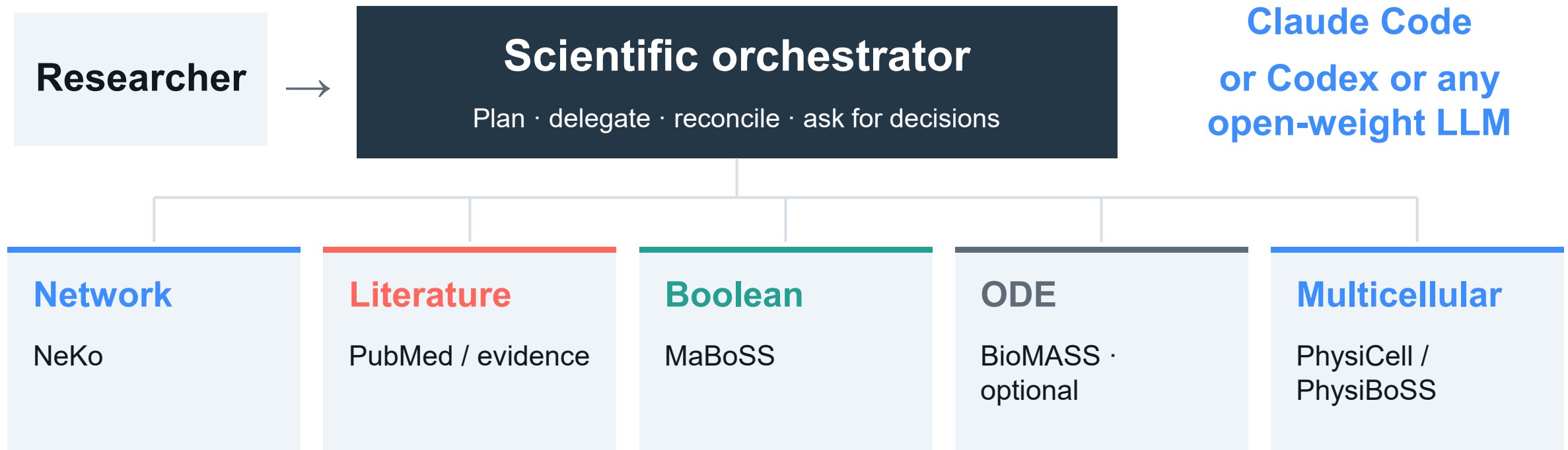
RESEARCHER

Define the question
Resolve scientific choices
Accept the limitations

The deliverable is a model plus the evidence, decisions and steps behind it.

One orchestrator coordinates five specialist roles

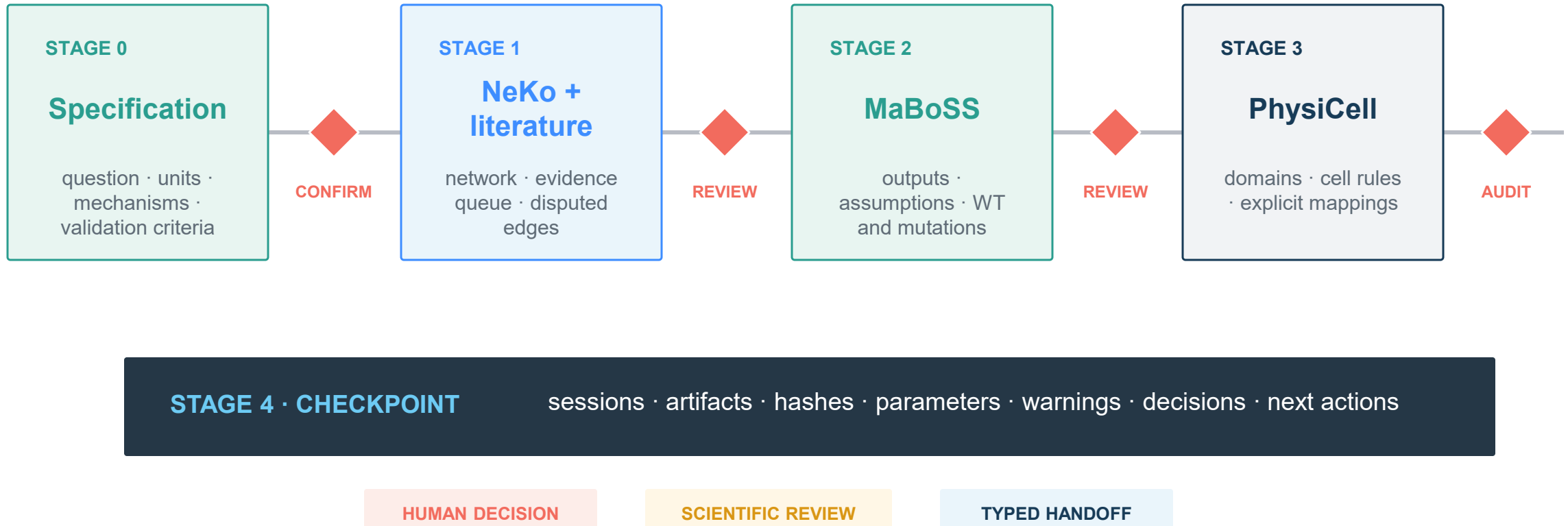
The researcher interacts with the orchestrator; each specialist receives a bounded task.



This case: network + literature → Boolean dynamics → multicellular configuration

Modelling alternates with review and approval

Every stage starts from verified state and ends with scientific review, approval and a recoverable checkpoint.



Shared state prevents biological drift

Data harmonization is preserved, not preprocessing hidden inside an agent conversation.



Assumption (example)

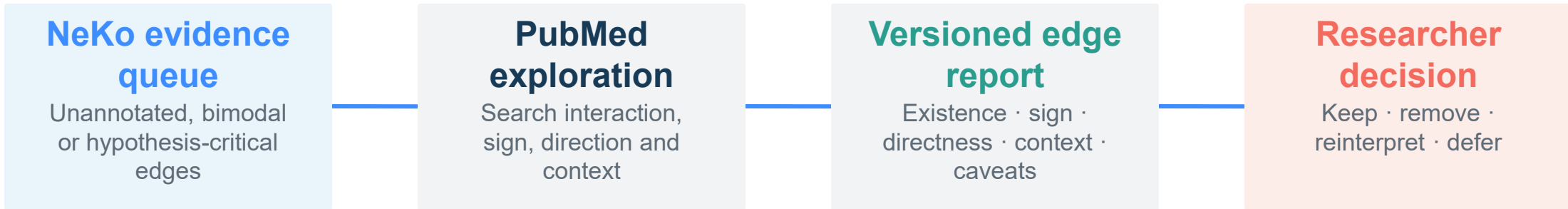
ID	Assumption	Evidence	Status	Consequence if false
A-001	ER+ luminal, RB1-proficient breast cancer is represented by ESR1 as a constant ON input feeding the core cell cycle.	Subtype choice by user, 2026-10-02	accepted (user)	Model represents a different context; V4b becomes uninformative.

Data dict (example)

Name	Type	Biological meaning	Unit	Source
ESR1	Boolean input	Oestrogen receptor- α activity (ER+ driver)	—	SIGNOR seed
CCND1	Boolean	Cyclin D1 (CycD)	—	SIGNOR seed

Literature review is wired into model decisions

A literature agent can investigate uncertain NeKo edges, but the researcher decides whether evidence should change the model.



CELL-CYCLE EVIDENCE

30

targeted edge reports

4

context-dependent verdicts

29 / 50

final edges with reports

The agent exposes a scientific decision

Example: an activating database edge does not match the reviewed biological context.

• Agent "Review candidate oscillator edges" finished · 4m 20s

← ☒ CDK1-APC/C ☐ CDK1-E2F1 ☐ CCND1-CDK6 ☒ Submit →

CDK1-E2F1 (activating in SIGNOR) is poorly supported: the activating evidence is in vitro only, and in vivo the kinase is mainly CycA-CDK2, which switches E2F1 off. What should happen to this edge?

- > 1. Remove; add CDK2-E2F1 (Recommended)
Drop CDK1-E2F1 and add CDK2-E2F1 (manual-literature, PMIDs 7969176, 7838523, 8548802 from the CDK1-E2F1 report). Adds the S-phase shut-off of E2F1 (a negative feedback via CycE-CDK2) and removes an unsupported positive loop.
- 2. Remove only
Drop CDK1-E2F1 and add nothing. E2F1 shut-off then depends only on RB1, which comes back on late.
- 3. Flip to CDK1-E2F1
Keep the edge but make it inhibitory. The sign then matches the evidence, but the kinase it names (CDK1 rather than CDK2) does not.
- 4. Type something.

5. Chat about this

The approved change is checked against the result

The system reports history, edge counts, exact differences and export readiness.

```
Read 2 files, called neko 12 times
• Import verified (31/113, no bimodal, references preserved). Applying the authorized
  removals now.
  Called neko 14 times
• History is linear, as expected (states 0 to 12). Now removing the two edges.
  Called neko 6 times
• Final edge list matches the expected 50 signed edges one-to-one, with no extras and no
  duplicates. Next I'll run the state comparison and the SIF export.
  Called neko 2 times
• The diff accounts for all 63 removed edges (113 - 50) and adds nothing. Exporting the SIF
  Called neko
• Verification passed on every criterion, so the authorized gated handoff export can go
  ahead.
  Called neko
```

Inspect

Read current state and history

Compare

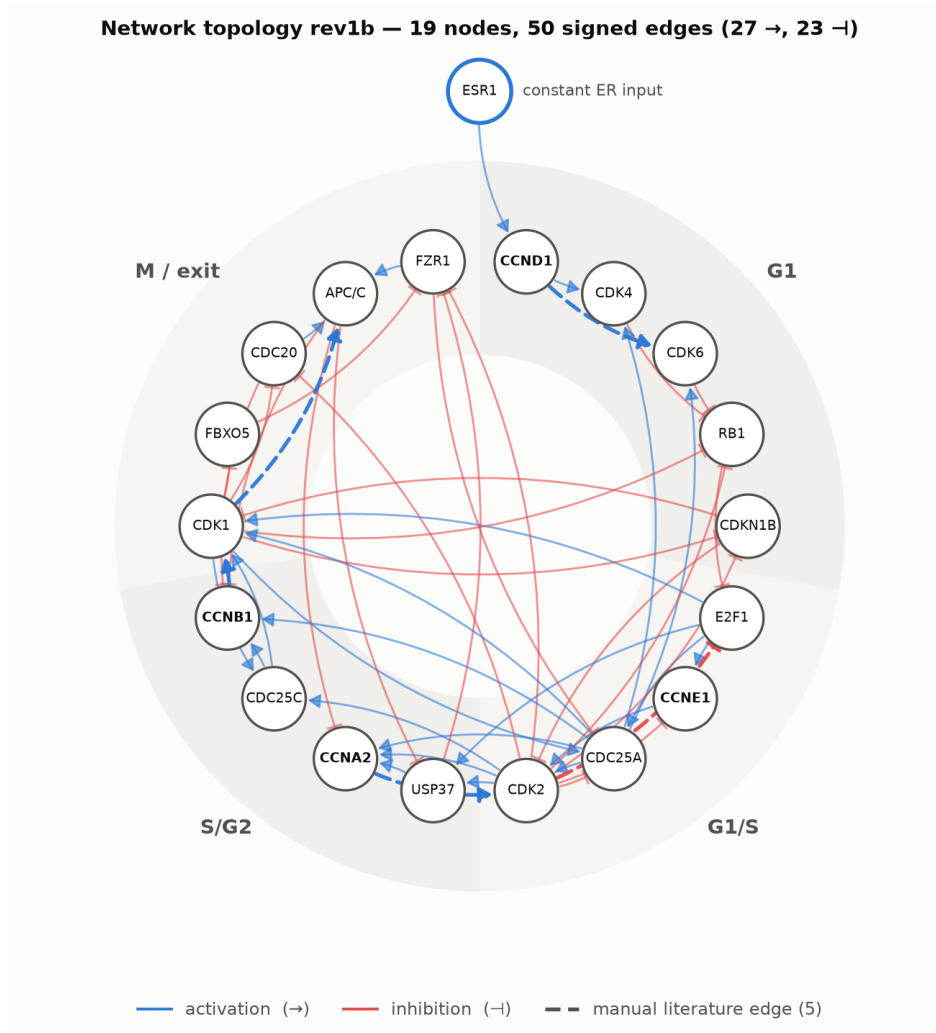
Match the approved edge set

Preserve

Export artifacts and retain
provenance

A small model with a complete modelling history

ER+ breast-cancer cell cycle: the toy system used to test the workflow.



31 → 19

nodes after curation

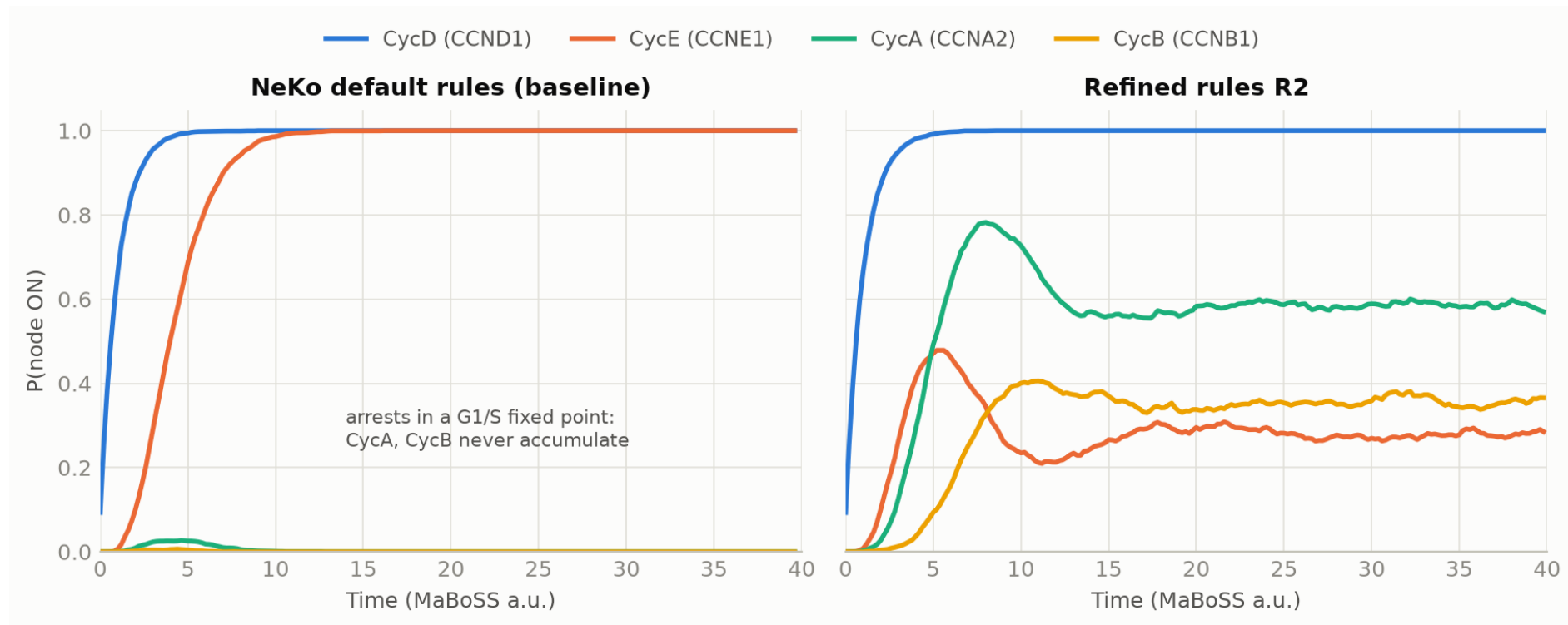
108 → 50

signed edges

Every revision is linked to its source session and decisions.

A plausible topology did not automatically cycle

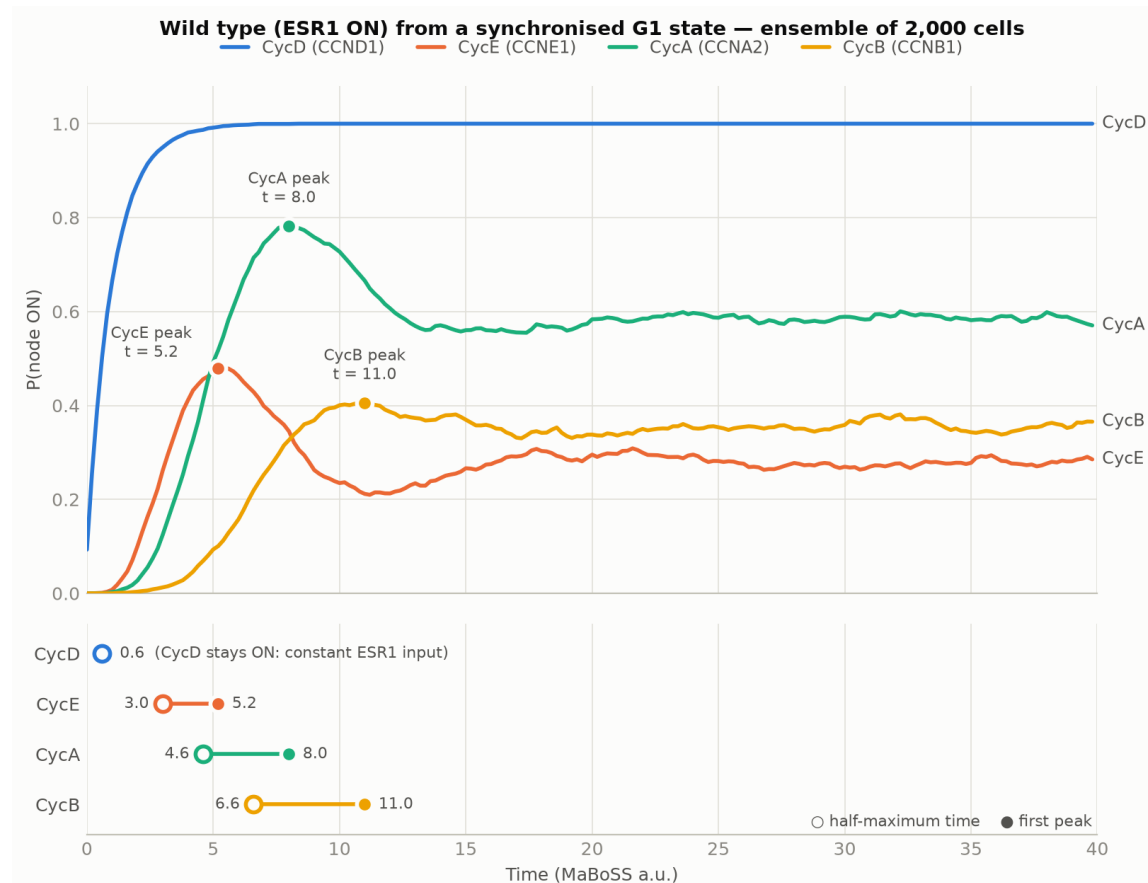
Default Boolean rules produced a G1/S lock. The system diagnosed it and tested refinements.



Baseline fails → R1 rejected → R2 accepted with limitations

The refined model recovers the first cyclin wave

Wild type, constant ESR1 input; 2,000 simulated cells from a synchronized G1 state.



D → E → A → B

First peaks

CycE 5.2 a.u.

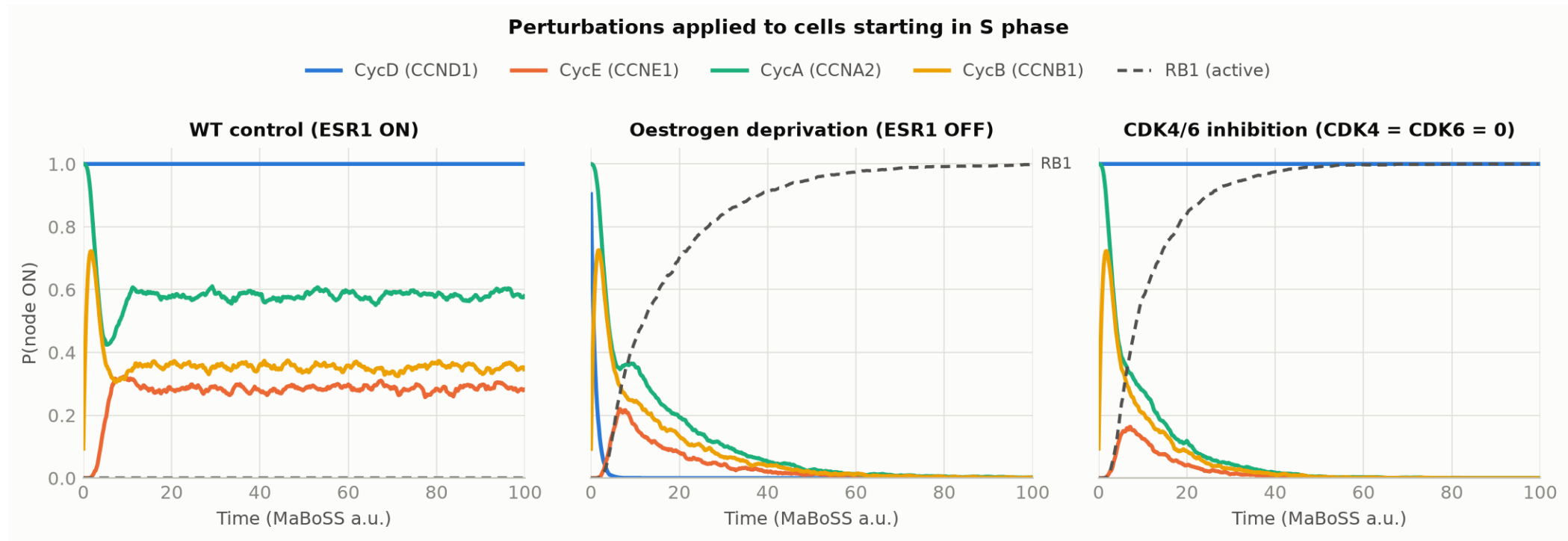
CycA 8.0 a.u.

CycB 11.0 a.u.

Population waves blur as individual cells lose synchrony.

Perturbations are tested from cells already cycling

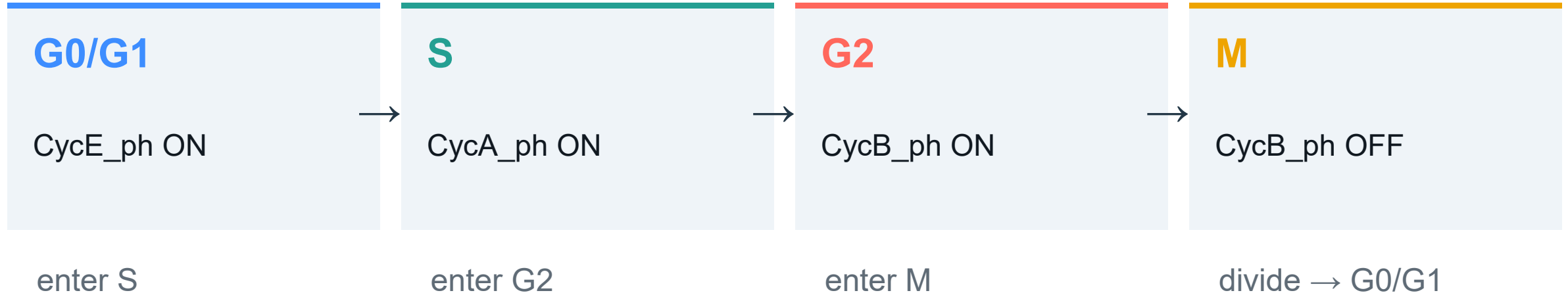
ESR1 OFF and CDK4/6 OFF lead to G1 arrest, including from S and G2/M starts.



WT continues; perturbed cells complete transient activity and then arrest.

From Boolean states to a multicellular configuration

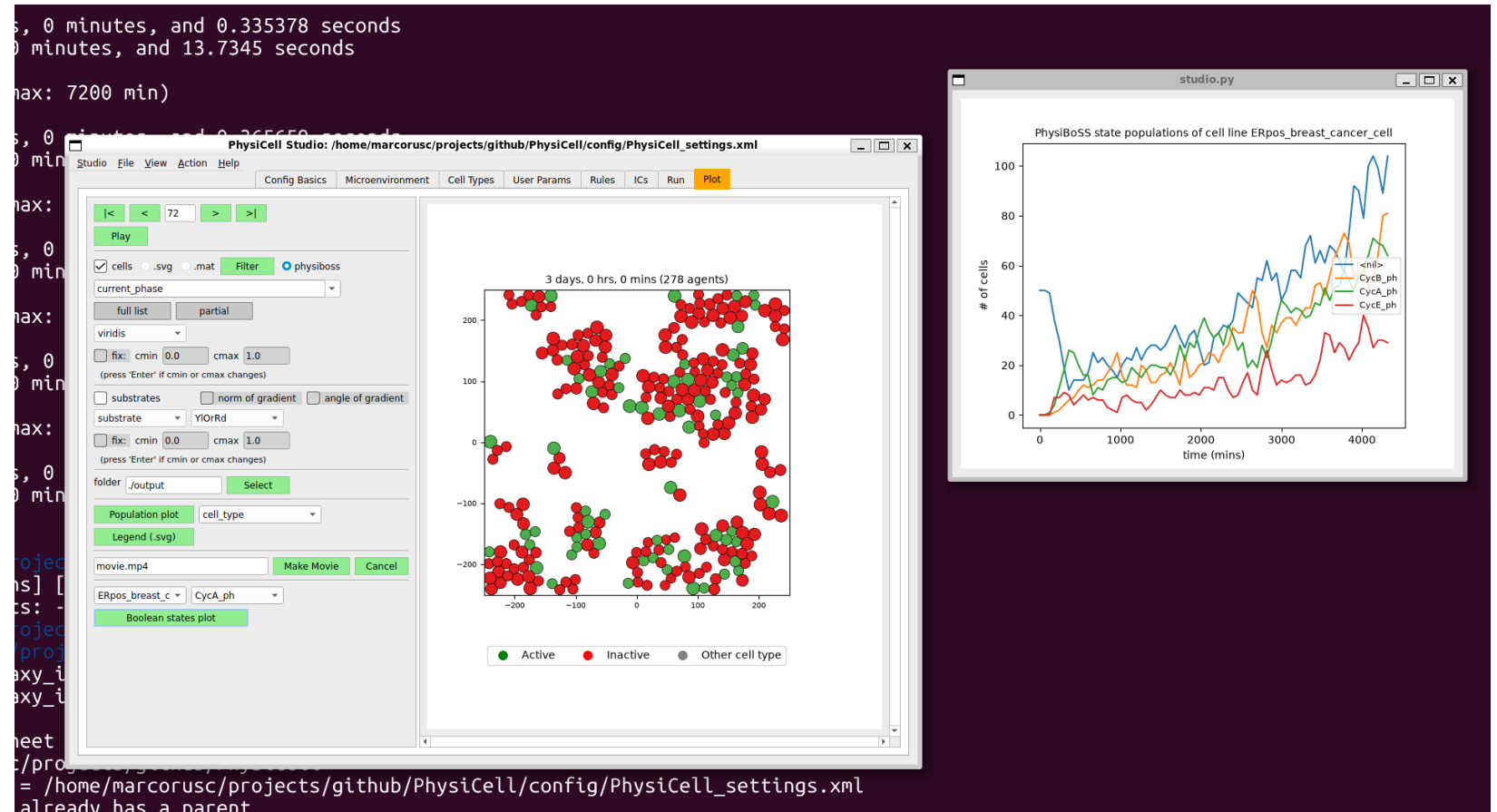
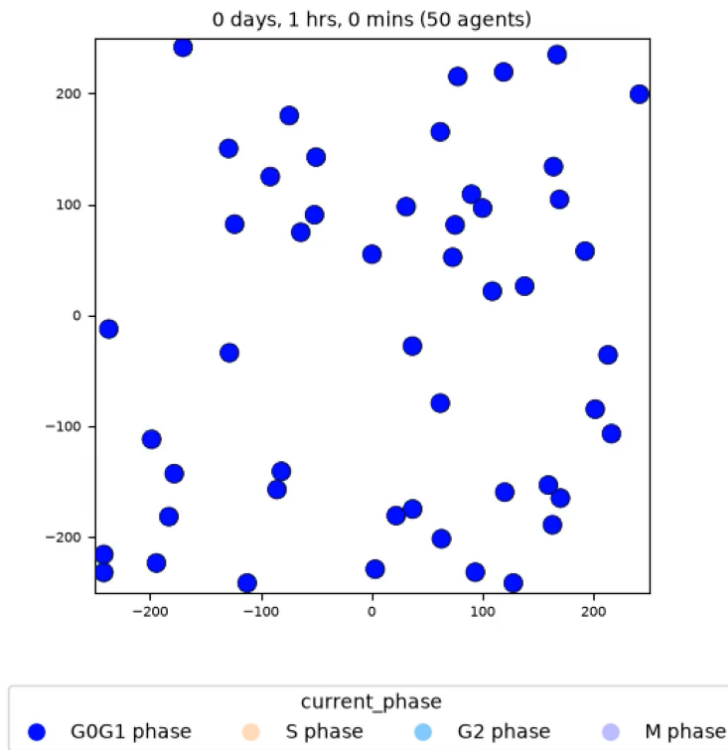
R3 adds three exclusive phase readouts while keeping the 19-rule R2 core unchanged.



Explicit mapping + explicit time-scale hypothesis + saved BND / CFG / XML

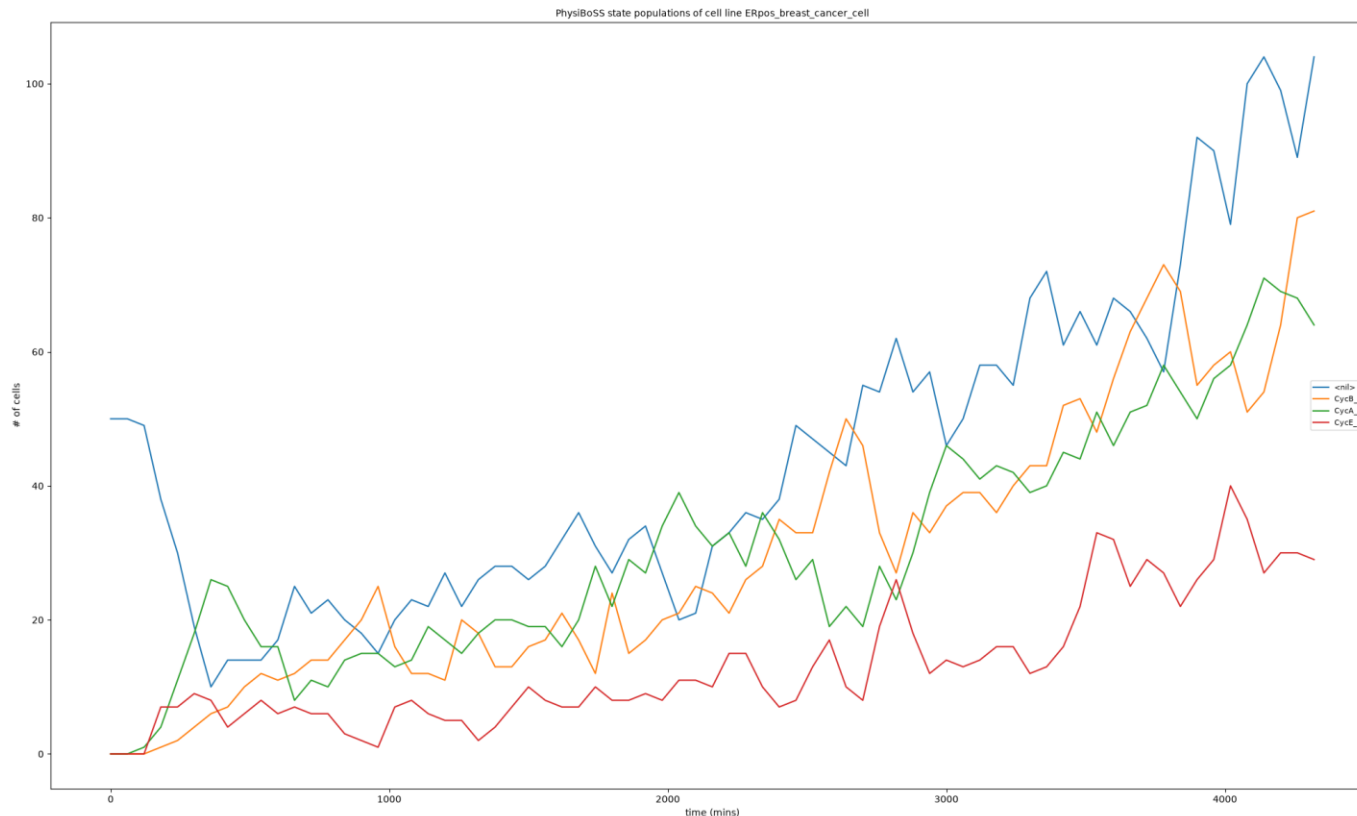
The model is visible at the multicellular scale

User-supplied PhysiCell Studio screenshot of the subsequent PhysiBoSS execution.



Intracellular states can be followed across the population

The same simulation exposes CycE_ph, CycA_ph and CycB_ph readout populations over time.



Boolean readouts

■ CycB_ph

■ CycA_ph

■ CycE_ph

■ <nil>: none ON

Cell counts over time.
Not measured phase
fractions.

Review made the result more defensible

The audit trail includes mistakes, rejected alternatives and repairs.

Provenance

Missing approval and rule tags were identified and repaired.

Validation

Arrest tests were extended to S and G2/M starts.

Reproducibility

Overlapping random seeds were flagged as non-independent.

R2 accepted with limitations: fitted rules, incomplete G1 reset, 32% canonical onsets.

Some stats directly from claude:

The recorded benchmark covers specification through reviewed Boolean dynamics.

87 min

NeKo → reviewed MaBoSS

30

edge evidence reports

1.36 M

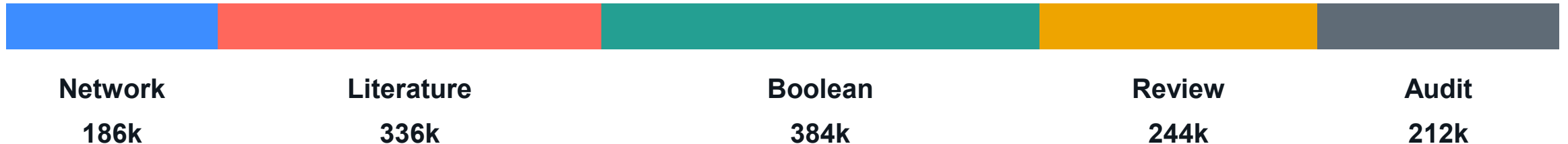
recorded subagent tokens

5

batches of researcher decisions

13

specialist invocations



LLM choice and coordination shape the workflow

The example demonstrates feasibility, not a benchmark across models or agent platforms.

LLM choice

Test tool use, scope adherence and recovery on a small representative task.
Compare quality, latency and token use.

Coordination

Context and handoffs add overhead.
Missing artifacts and failed sessions can trigger repair and repeated work.

Support for Claude Code and Codex does not establish equal behaviour or performance.

Multi-agent systems for reproducible biology

A prompt-driven workflow can produce a model and a record that another researcher can examine.

Prompt

State the question and validation targets.

Decide

Inspect evidence and approve scientific choices.

Reproduce

Recover inputs, rules, settings, results and rationale.

NeKo → MaBoSS → PhysiBoSS, with researcher judgment at every consequential step.

Explore the system and the modelling tools

Two complementary repositories: the research workflow and the MCP interfaces it uses.

Agentic system for modelling

Orchestrator, specialist roles, skills and scientific records

https://github.com/marcorusc/agentic_system_for_modelling

MCP bio-modelling servers

Tool interfaces for NeKo, MaBoSS and PhysiCell / PhysiBoSS

<https://github.com/marcorusc/mcp-biomodelling-servers>

[Related article: pubmed.ncbi.nlm.nih.gov/42297831/](https://pubmed.ncbi.nlm.nih.gov/42297831/)

Thank you

Questions?

With thanks to Anthropic

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And to the developers and communities behind NeKo, MaBoSS and PhysiBoSS.